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CE F231 Fluid Mechanics

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Lecture 19: Fluid Kinematics

Recap: Stream Function (Ψ - psi)

- Scalar function; defined for 2D flow
- Partial derivative of Ψ with respect to any direction (x or y) gives velocity component at right angle to that direction
- Satisfies continuity eqn. for 2D incompressible flow

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0.$$

$$\frac{\partial \Psi}{\partial x} = v$$

$$\frac{\partial \Psi}{\partial y} = -u$$

OR

$$v = -\frac{\partial \Psi}{\partial x}$$

$$u = \frac{\partial \Psi}{\partial y}$$

Recap: Stream Function (Ψ - psi)

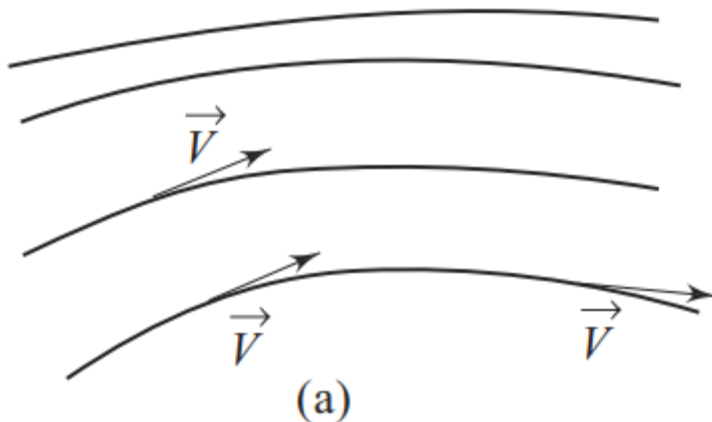
$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v dx + u dy$$

$\psi = \text{Constant}$
 $d\psi = 0$ (Equi-stream function line)

$$v = -\frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial y}$$

Streamlines

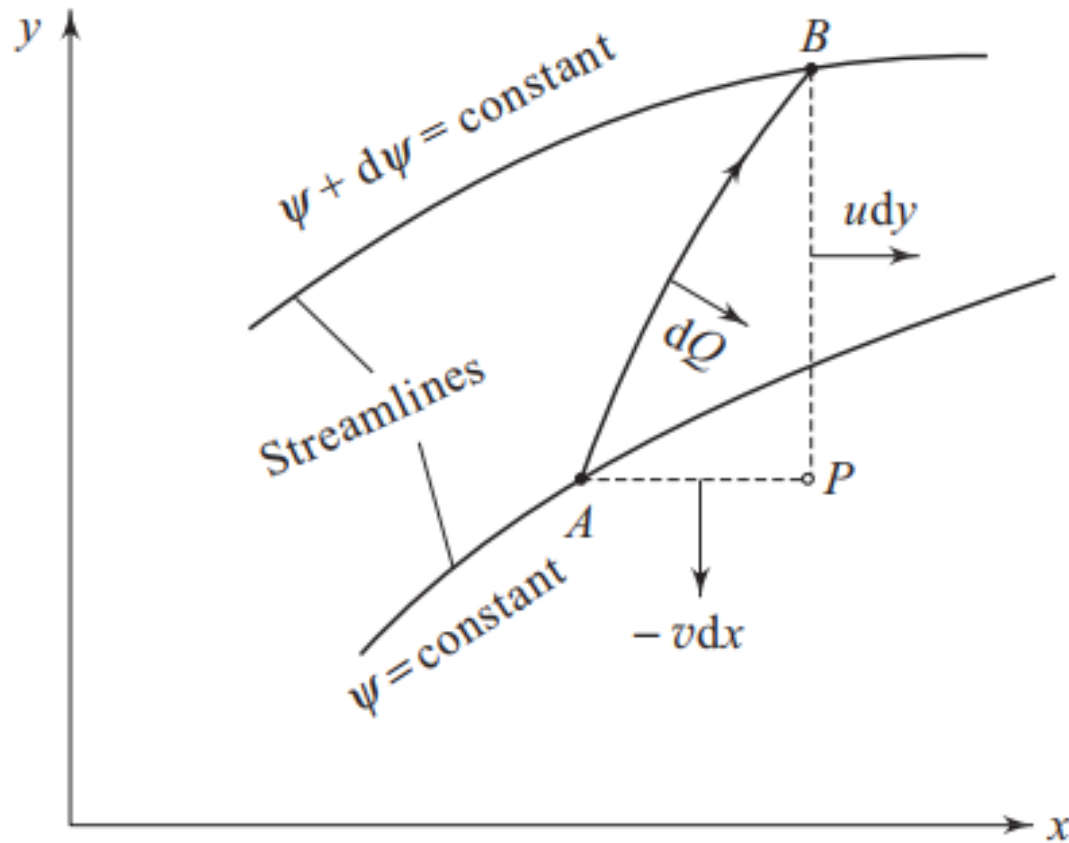


$$\vec{V} \times d\vec{S} = 0$$

$$\frac{dx}{u} = \frac{dy}{v}$$

Value of stream function does not change along a streamline

Recap: Physical Significance of Stream Function



$$dQ = u dy - v dx$$

$$u = \frac{\partial \psi}{\partial y} \quad v = -\frac{\partial \psi}{\partial x}$$

$$dQ = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy$$

Flow rate through any line joining two points on two streamlines = difference between the respective values of Ψ

Recap: Stream Function (Ψ - psi)

➤ For 2D, incompressible, and irrotational flow

$$\text{Vorticity} = 2 \times \text{Rotation} = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} = 0$$

$$v = -\frac{\partial \psi}{\partial x}$$

$$\frac{\partial}{\partial x} \left(-\frac{\partial \psi}{\partial x} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \psi}{\partial y} \right) = 0$$

$$u = \frac{\partial \psi}{\partial y}$$

$$\nabla^2 \psi = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = 0$$

(Laplace equation)

Velocity Potential Function (Φ - phi)

- Scalar function of space and time
- Negative partial derivative of Φ with respect to any direction (x, y, or z) gives velocity component in that direction

$$\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z} = 0.$$

$$\frac{\partial}{\partial x} \left(-\frac{\partial \Phi}{\partial x} \right) + \frac{\partial}{\partial y} \left(-\frac{\partial \Phi}{\partial y} \right) + \frac{\partial}{\partial z} \left(-\frac{\partial \Phi}{\partial z} \right) = 0$$

$$\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0.$$

$$u = -\frac{\partial \Phi}{\partial x}$$

$$v = -\frac{\partial \Phi}{\partial y}$$

$$w = -\frac{\partial \Phi}{\partial z}$$

If Φ satisfies Laplace equation, then it will represent some type of incompressible fluid flow

Velocity Potential Function (Φ - phi)

$$\omega_x = \frac{1}{2} \left(\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z} \right) \quad \omega_y = \frac{1}{2} \left(\frac{\partial u}{\partial z} - \frac{\partial w}{\partial x} \right) \quad \omega_z = \frac{1}{2} \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right)$$

$$\omega_z = \frac{1}{2} \left[\frac{\partial}{\partial x} \left(-\frac{\partial \phi}{\partial y} \right) - \frac{\partial}{\partial y} \left(-\frac{\partial \phi}{\partial x} \right) \right] = \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial x \partial y} + \frac{\partial^2 \phi}{\partial y \partial x} \right]$$

$$\omega_y = \frac{1}{2} \left[\frac{\partial}{\partial z} \left(-\frac{\partial \phi}{\partial x} \right) - \frac{\partial}{\partial x} \left(-\frac{\partial \phi}{\partial z} \right) \right] = \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial z \partial x} + \frac{\partial^2 \phi}{\partial x \partial z} \right]$$

$$\omega_x = \frac{1}{2} \left[\frac{\partial}{\partial y} \left(-\frac{\partial \phi}{\partial z} \right) - \frac{\partial}{\partial z} \left(-\frac{\partial \phi}{\partial y} \right) \right] = \frac{1}{2} \left[-\frac{\partial^2 \phi}{\partial y \partial z} + \frac{\partial^2 \phi}{\partial z \partial y} \right]$$

$$\omega_z = \omega_y = \omega_x = 0.$$

$$u = -\frac{\partial \phi}{\partial x}$$

$$v = -\frac{\partial \phi}{\partial y}$$

$$w = -\frac{\partial \phi}{\partial z}$$

If Φ exists, flow must be irrotational



Velocity Potential Function (Φ - phi)

There is a **vector identity** (prove it for yourself!) that states for any scalar, ϕ ,

$$\nabla \times \nabla \phi = 0 \quad (\text{Curl of gradient is 0})$$

By definition, for *irrotational flow*,

$$\nabla \times \vec{V} = 0$$

Therefore

$$\vec{V} = \nabla \phi \quad (\text{Velocity vector can be defined as gradient of scalar function})$$

where $\phi = \phi(x, y, z, t)$ is the velocity *potential function*. Such that the components of velocity in Cartesian coordinates, as functions of space and time, are

$$u = \frac{\partial \phi}{\partial x}, \quad v = \frac{\partial \phi}{\partial y} \quad \text{and} \quad w = \frac{\partial \phi}{\partial z}$$

Some authors define velocity components as positive partial derivatives of Φ wrt respective directions

Velocity Potential Function (Φ - phi)

$$d\phi = \frac{\partial \phi}{\partial x} dx + \frac{\partial \phi}{\partial y} dy$$

$$= - u dx - v dy$$

$$= - (u dx + v dy).$$

$$d\phi = 0 \quad (\text{Equipotential line})$$

$$- (u dx + v dy) = 0 \text{ or } u dx + v dy = 0$$

$$\frac{dy}{dx} = - \frac{u}{v}$$

Flow Net

For equi-stream function lines

$$d\psi = \frac{\partial \psi}{\partial x} dx + \frac{\partial \psi}{\partial y} dy = -v dx + u dy = 0$$

$$\frac{dy}{dx} = \frac{v}{u}$$

For equipotential lines

$$d\phi = -(u dx + v dy) = 0 \text{ or } u dx + v dy = 0$$

$$\frac{dy}{dx} = -\frac{u}{v}$$

Flow Net

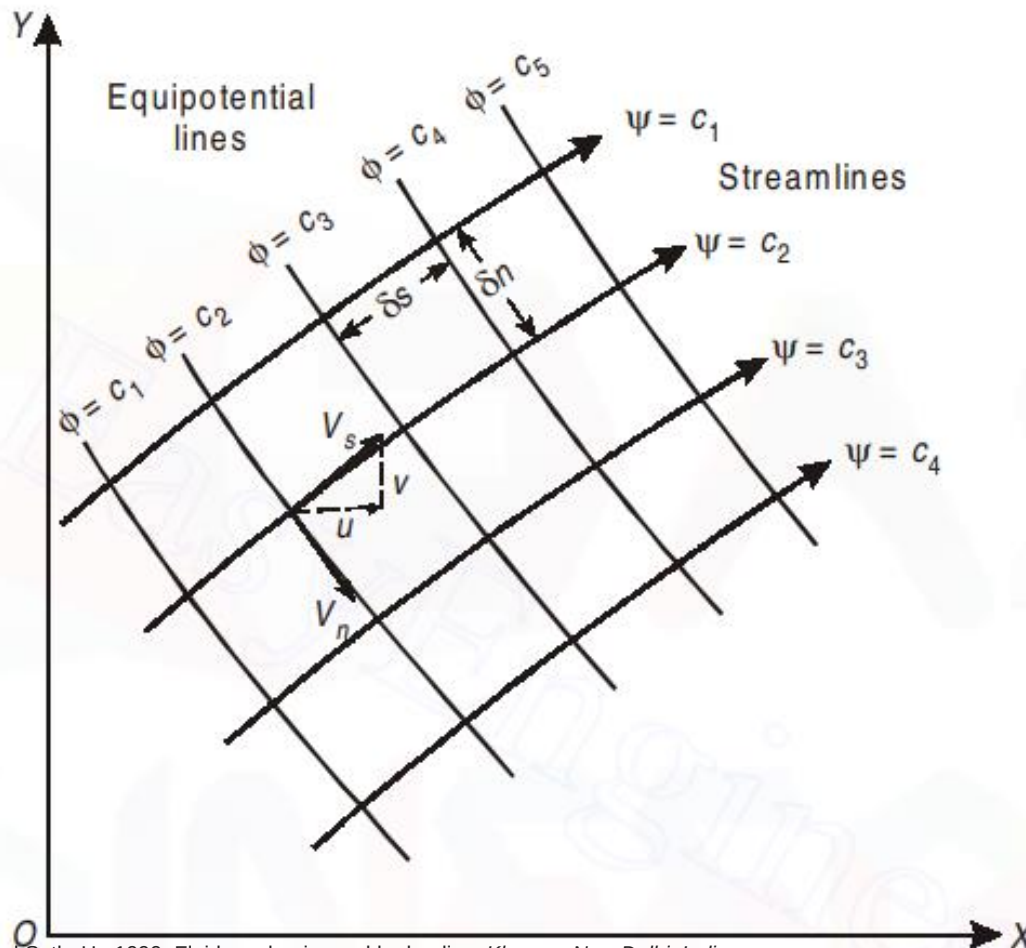
An important tool for analyzing 2D irrotational flow problems

Partial derivative of Ψ with respect to any direction gives velocity component at right angle to that direction

$$\frac{\partial \psi}{\partial s} = V_n = 0$$

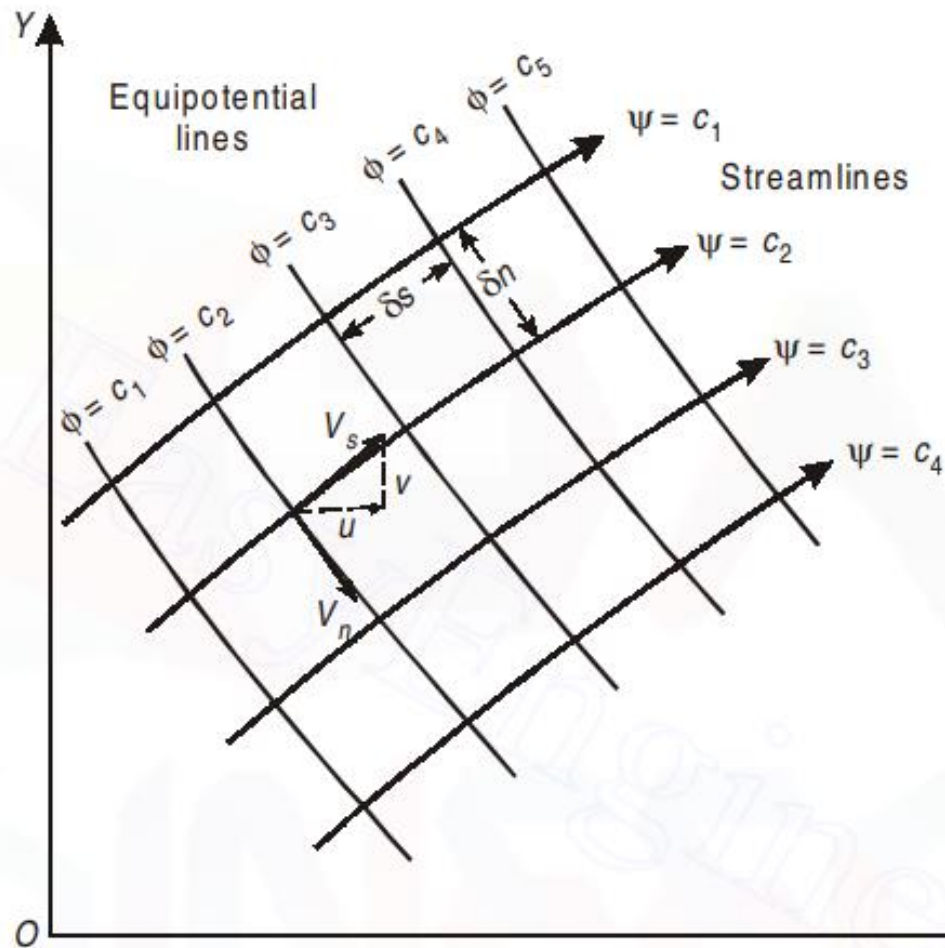
$$\frac{\partial \psi}{\partial n} = -V_s$$

$$C_2 - C_1 = ?$$



Flow Net

An important tool for analyzing 2D irrotational flow problems



Negative partial derivative of Φ with respect to any direction gives velocity component in that direction

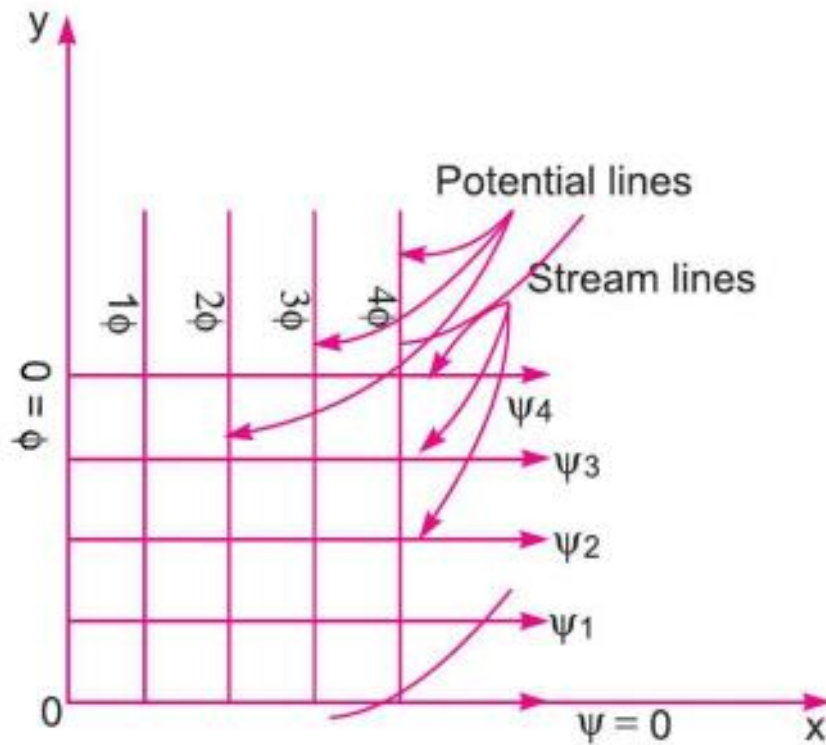
$$-\frac{\partial \phi}{\partial n} = V_n$$

$$V_n = 0$$

$$-\frac{\partial \phi}{\partial s} = V_s$$

Flow Net

Flow in X-direction through horizontal pipe of uniform diameter (Uniform Flow)



$$u = - \frac{\partial \phi}{\partial x}$$

$$v = - \frac{\partial \phi}{\partial y}$$

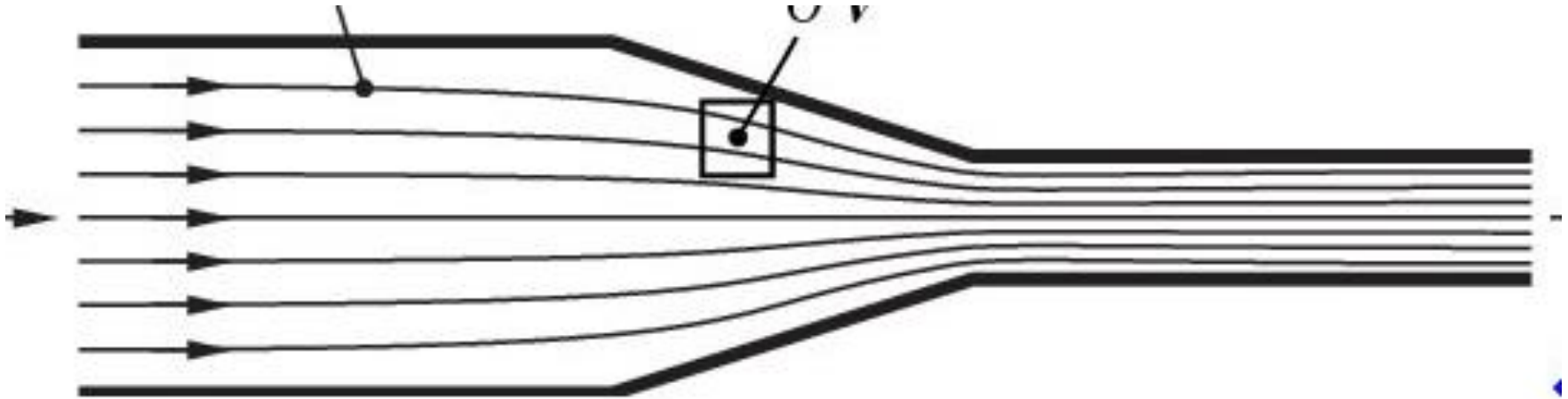
$$v = - \frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial y}$$

All equipotential lines will be parallel and equally spaced

Flow Net

Flow through converging pipe (Non-uniform Flow)



u only; both u and v ?

Expected variation of u ?

u and v from ψ for 3 regions?

Based on u – predict ϕ

Spacing between equipotential lines

Equipotential lines shape in 3 regions

$$u = - \frac{\partial \phi}{\partial x}$$

$$v = - \frac{\partial \phi}{\partial y}$$

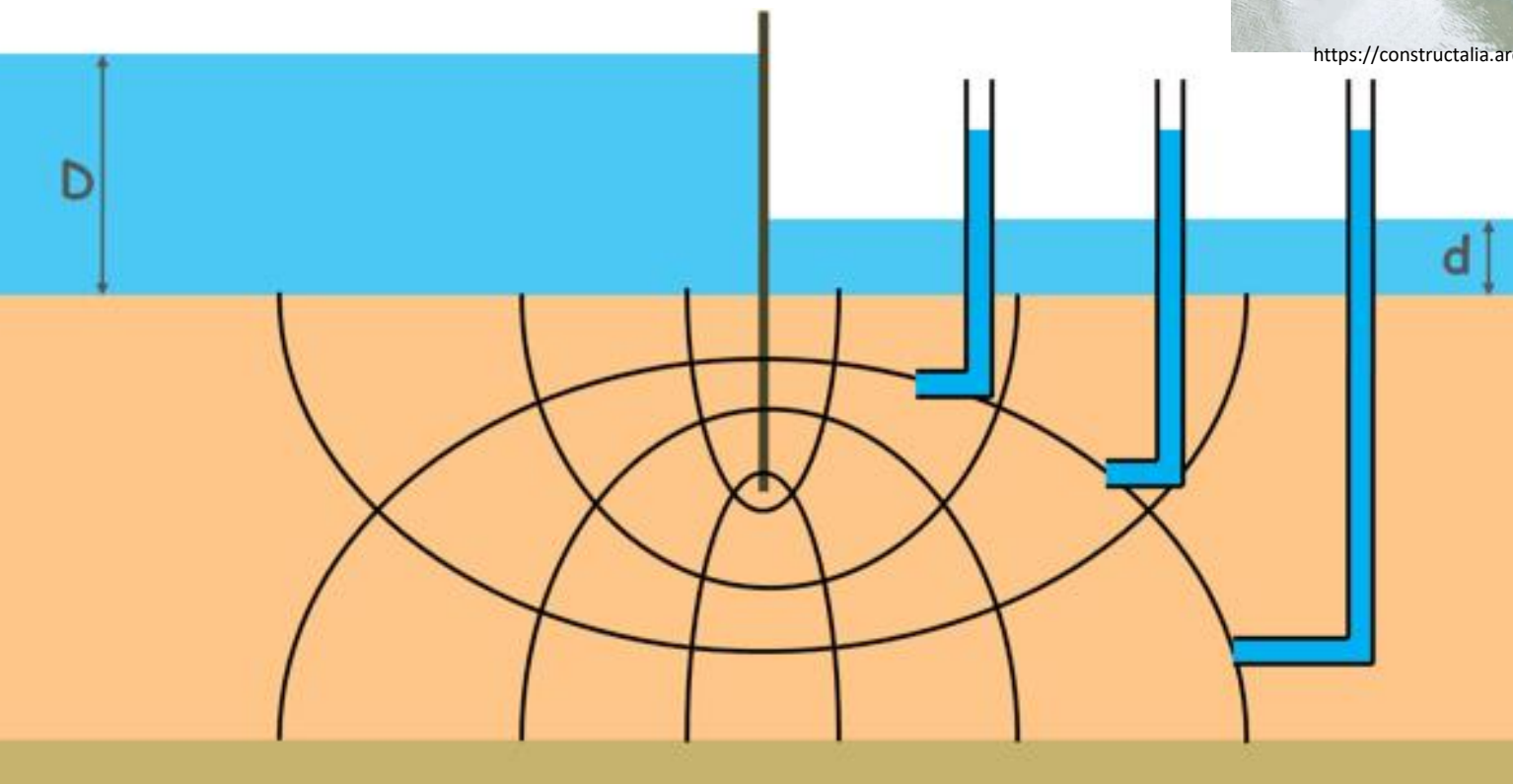
$$v = - \frac{\partial \psi}{\partial x}$$

$$u = \frac{\partial \psi}{\partial y}$$

Flow Net



https://constructalia.arcelormittal.com/en/products/sheet_piles







THANK YOU